EAST WEST INSTITUTE OF TECHNOLOGY

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

Technical Seminar Presentation
on

ENABLING INDUSTRIAL INTERNET OF THINGS: EDGE-TO-CLOUD COMPUTING

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ABSTRACT

The Industrial Internet of Things (IIoT) enables automation, efficiency, and intelligent decision-making through real-time data collection and analysis. However, traditional cloud-centric IIoT architectures face challenges such as managing large data volumes, limited edge bandwidth, and concerns over data confidentiality. Distributed edge-to-cloud computing emerges as a potential solution, also paving the ground for edge-to-cloud data analytics and distributed Artificial Intelligence (AI) to obtain insights for decision-making and predictive maintenance. Despite the potential, however, there is a lack of comprehensive studies identifying key requirements for distributed edge-to-cloud IIoT and analyzing to what extent merging IoT platforms meet those requirements. The scope of this article is to survey existing literature to identify key requirements in IIoT from the perspective of distributed edge-to-cloud computing. We provide a comparative analysis of three prominent IoT platforms, namely ThingsBoard, Eclipse Ditto, and Microsoft Azure IoT, and assess how these platforms meet the key IIoT requirements. Finally, we identify open challenges and potential research opportunities based on the insights gained from the analysis of the three IoT platforms, thereby setting the stage for future work.

INTRODUCTION

- The IIoT integrates smart devices, sensors, and control systems to collect, share, and analyze industrial data. The synergy of edge and cloud computing ensures real-time processing and long-term insights, vital for modern industry operations
- By enabling machines to communicate and make data-driven decisions, IIoT transforms traditional factories into intelligent, self-optimizing systems. This shift improves productivity, reduces downtime, and enhances safety in industrial environments.
- Continuously monitors machine health, production status, and environmental conditions.
- Critical for time-sensitive industrial operations and safety systems.
- Enables predictive maintenance, process optimization, and autonomous decision-making.
- Combines fast local processing (edge) with scalable data analytics (cloud) for better performance.

MOTIVATION

The rapid digital transformation of industries demands faster, smarter, and more responsive systems. Traditional cloud computing, though powerful, struggles with latency and bandwidth limitations in time-sensitive environments. This has fueled the need for edge computing to process data closer to the source, enabling real-time actions. Industries today aim to enhance automation, reduce downtime, and achieve predictive maintenance. By combining edge and cloud computing, IIoT systems can deliver high reliability, real-time analytics, and scalable intelligence. This hybrid model ensures industrial operations are not just connected but intelligent, efficient, and future-ready.

The integration of IIoT with edge-to-cloud computing enables continuous data collection and advanced analytics, fostering innovation in product design, quality control, and supply chain management. This helps industries stay competitive in a fast-evolving digital landscape.

PROBLEM STATEMENT

Centralized computing models face significant challenges when applied to Industrial Internet of Things (IIoT) environments. These systems often introduce high latency, making them unsuitable for real-time industrial applications where immediate response is crucial. As the number of connected devices increases, scalability becomes difficult to manage, and the burden on bandwidth and processing grows. Additionally, centralized architectures raise concerns over data security and privacy, especially when sensitive industrial information is transmitted to the cloud. While edge computing offers low-latency processing close to data sources, relying solely on the edge can limit large-scale data analytics and long-term storage. Therefore, a hybrid edge-to-cloud model is necessary to balance local responsiveness with cloud-based intelligence. However, integrating these layers introduces complexities in data synchronization, system orchestration, and overall infrastructure management. This creates a pressing need for innovative solutions to manage distributed IIoT systems efficiently and securely.

HISTORY

- 1980s–1990s:
  - Systems were isolated with minimal remote data access.
- Early 2000s:
  - Introduction of industrial automation and sensors.
  - Basic data logging, but lacked real-time analytics.
- 2010–2015:
  - Cloud computing gained popularity for big data and storage.
  - enabling device connectivity and remote monitoring.
- 2015–2020:
  - Rise of edge computing for latency-sensitive processing.
  - Need for real-time insights drove edge adoption in industries.
- 2020–2024:
  - Hybrid edge-to-cloud models implemented widely. Focus on smart factories,
  - AI at the edge, and system orchestration. Emphasis on cybersecurity, data privacy, and reliable connectivity.

FEATURES

- Real-Time Processing: Enables immediate responses to critical events through local edge devices.
- Low Latency: Reduces data transmission delays, crucial for time-sensitive industrial operations.
- Scalability: Supports expansion from local edge nodes to global cloud platforms.
- Hybrid Architecture: Combines strengths of both edge (speed) and cloud (powerful analytics).
- Improved Reliability: Local processing ensures system functionality even during network interruptions.
- Enhanced Security: Keeps sensitive data at the edge, reducing risks from cloud breaches

WORKING MECHANISM

- In an IIoT system, data is first collected through sensors embedded in industrial devices.
- This data is immediately processed by edge devices located close to the source, allowing for real-time decisions and fast responses.
- Only critical or summarized data is transmitted to the cloud, where it is stored, analyzed, and used for generating deeper insights over time.
- This hybrid edge-to-cloud structure ensures a balance between processing speed and scalability. By minimizing the volume of data sent to the cloud, the system reduces bandwidth usage and latency. Moreover, it enhances the accuracy and timeliness of decisions made within industrial operations.
- A continuous feedback loop between edge devices, cloud systems, and actuators helps maintain efficiency and adapt to real-time conditions.

COMPARATIVE STUDY

A comparative analysis of cloud computing, edge computing, and the integrated edge-to-cloud model highlights key differences and advantages. Cloud computing offers massive scalability, centralized storage, and high computational power but suffers from latency and dependence on stable internet connectivity. Edge computing, on the other hand, processes data locally, offering real-time responsiveness and reduced bandwidth usage, though it may face scalability limitations. The edge-to-cloud architecture merges the strengths of both models—handling immediate processing at the edge and utilizing the cloud for large-scale analytics and storage. This hybrid setup ensures optimized performance, lower latency, higher reliability, and better data orchestration across industrial systems

PROS AND CONS

Pros:

- Quicker decision-making with local edge processing
- Reduced network bandwidth consumption
- Supports automation and AI applications
- Improved system reliability and responsiveness
- Enables predictive maintenance and fault detection

Cons:

- Complex integration and system management
- Data consistency and synchronization challenges
- Higher initial infrastructure and deployment costs
- Requires skilled personnel for configuration and maintenance
- Security management across multiple layers is demanding

APPLICATIONS

- Manufacturing: Predictive maintenance, real-time equipment monitoring
- Energy Sector: Smart grids, demand forecasting, energy usage optimization
- Healthcare: Remote patient monitoring, diagnostics via edge devices
- Logistics: Real-time supply chain tracking, smart inventory systems
- Agriculture: Smart farming, environmental monitoring, precision irrigation
- Retail: Customer behavior analysis, stock management
- Smart Cities: Traffic control, public safety, environmental data collection

FUTURE TRENDS

- AI/ML at the Edge: Enables autonomous decision-making and real-time intelligence
- 5G/6G Integration: Ultra-low latency networks for faster, more reliable communication
- Open-Source : Simplifies deployment and management of hybrid systems
- Energy-Efficient Computing: Focus on sustainable, low-power edge devices
- Standardized Frameworks: Improved interoperability and security across platforms
- Increased Cybersecurity Measures: To protect distributed IIoT infrastructures
- Edge-Native Applications: Designed specifically for real-time industrial operations

CONCLUSION

The integration of edge-to-cloud computing within Industrial Internet of Things (IIoT) environments marks a significant leap toward smarter, faster, and more efficient industrial operations. This hybrid model ensures low latency, real-time responsiveness at the edge, and powerful data analytics through the cloud. While challenges like data management, synchronization, and integration still exist, ongoing technological advancements and standardization efforts are steadily addressing them. As industries continue to evolve, edge-to-cloud computing will play a pivotal role in driving automation, enhancing productivity, and enabling intelligent decision-making across sectors.

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Thank You